

WeBWork assignment number WW_02 is due : 07/18/2008 at 11:00pm CDT.

The

(* replace with url for the course home page *)

for the course contains the syllabus, grading policy and other information.

This file is /conf/snippets/setHeader.pg you can use it as a model for creating files which introduce each problem set.

The primary purpose of WeBWork is to let you know that you are getting the correct answer or to alert you if you are making some kind of mistake. Usually you can attempt a problem as many times as you want before the due date. However, if you are having trouble figuring out your error, you should consult the book, or ask a fellow student, one of the TA's or your professor for help. Don't spend a lot of time guessing – it's not very efficient or effective.

Give 4 or 5 significant digits for (floating point) numerical answers. For most problems when entering numerical answers, you can if you wish enter elementary expressions such as $2 \wedge 3$ instead of 8, $\sin(3 * \pi/2)$ instead of -1, $e \wedge (\ln(2))$ instead of 2, $(2 + \tan(3)) * (4 - \sin(5)) \wedge 6 - 7/8$ instead of 27620.3413, etc. Here's the **list of the functions** which WeBWork understands.

You can use the Feedback button on each problem page to send e-mail to the professors.

1. (1 pt) ttulib/M2360/chap2/leon.2.1/n01.pg

(S. 2-I, cf. no. 1))

Given $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 3 \\ 1 & 5 & 2 \end{bmatrix}$

(a) Find the values of:

$\det(M_{21})$ _____

$\det(M_{22})$ _____

$\det(M_{23})$ _____

(b) Find the values of

A_{21} _____

A_{22} _____

A_{23} _____

(c) Use your answers from part (b) to compute $\det(A)$:

2. (1 pt) ttulib/M2360/chap2/leon.2.1/n02.pg

(S. 2-I, cf. no. 2))

Use determinants to determine whether the following 2×2 matrices are nonsingular:

(a) $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$, determinant: _____,

- singular
- nonsingular

(b) $\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}$, determinant: _____,

- singular
- nonsingular

3. (1 pt) ttulib/M2360/chap2/leon.2.1/n03.pg

(S. 2-I, cf. no. 3))

Evaluate the following determinants:

(a) $\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} =$ _____

(b) $\begin{vmatrix} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & 3 & 1 \end{vmatrix} =$ _____

(c) $\begin{vmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 4 & 2 & -4 \end{vmatrix} =$ _____

(d) $\begin{vmatrix} 1 & 0 & 1 & 3 \\ 2 & 1 & 1 & 5 \\ -2 & 0 & 1 & -1 \\ 3 & 1 & -1 & 0 \end{vmatrix} =$ _____

4. (1 pt) ttulib/M2360/chap2/leon.2.1/n05.pg

(S. 2-I, cf. no. 5))

Evaluate the following determinant. Write your answer as a polynomial in x .

$$\det \left(\begin{bmatrix} a+x & -b & c \\ 1 & x & 0 \\ 0 & 1 & x \end{bmatrix} \right) =$$

5. (1 pt) ttulib/M2360/chap2/leon.2.1/n06.pg

(S. 2-I, cf. no. 6))

Find all values of λ for which the following determinant will equal 0:

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix}$$

$\lambda =$ _____

(Write your answer as a list of numbers separated by commas.)

6. (1 pt) ttulib/M2360/chap2/leon.2.1/ur_la_6.2.pg

If $A = \begin{bmatrix} 0 & 9 \\ -8 & 4 \end{bmatrix}$,

then $\det(A) = \underline{\hspace{2cm}}$ and $A^{-1} = \begin{bmatrix} \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \end{bmatrix}$.

7. (1 pt) ttulib/M2360/chap2/leon.2.1/ur_la_6.4.pg

Find the determinant of the matrix

$$B = \begin{bmatrix} -3 & 4 & 0 \\ 5 & 3 & 3 \\ 4 & 0 & -1 \end{bmatrix}.$$

$\det(B) = \underline{\hspace{2cm}}$.

8. (1 pt) ttulib/M2360/chap2/leon.2.1/ur_la_6.5.pg

Find the determinant of the matrix

$$C = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 0 & 2 & 2 \\ 1 & 3 & -1 & 3 \\ -1 & 3 & 3 & -1 \end{bmatrix}.$$

$\det(C) = \underline{\hspace{2cm}}$.

9. (1 pt) ttulib/M2360/chap2/leon.2.2/n01.pg

(S. 2-2, cf. no. 1))

Evaluate each of the following determinants by inspection:

(a) $\begin{vmatrix} 0 & 2 & 0 \\ 1 & 1 & 1 \\ 2 & 3 & 8 \end{vmatrix} = \underline{\hspace{2cm}}$

(b) $\begin{vmatrix} 2 & 2 & 2 & 1 \\ 0 & 2 & 3 & 0 \\ 0 & 0 & 2 & -1 \\ -2 & -2 & -2 & 5 \end{vmatrix} = \underline{\hspace{2cm}}$

(c) $\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{vmatrix} = \underline{\hspace{2cm}}$

10. (1 pt) ttulib/M2360/chap2/leon.2.2/n03.pg

(S. 2-3, cf. no. 3))

For each of the following, compute the determinant and state whether the matrix is singular or nonsingular.

(a) $A = \begin{bmatrix} -3 & 1 \\ 9 & -3 \end{bmatrix}$, $\det(A) = \underline{\hspace{2cm}}$

- singular
- nonsingular

(b) $B = \begin{bmatrix} -3 & 1 \\ 9 & 3 \end{bmatrix}$, $\det(B) = \underline{\hspace{2cm}}$

- singular
- nonsingular

(c) $C = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $\det(C) = \underline{\hspace{2cm}}$

- singular
- nonsingular

(d) $D = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 4 & 2 \\ 2 & -4 & -8 \end{bmatrix}$, $\det(D) = \underline{\hspace{2cm}}$

- singular
- nonsingular

11. (1 pt) ttulib/M2360/chap2/leon.2.2/n04.pg

(S. 2-2, cf. no. 4))

Find all possible values of c that would make the following matrix singular.

$$\begin{bmatrix} c-1 & -2 & -3 \\ c-1 & c+1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

$c = \underline{\hspace{2cm}}$ (A list of numbers $c_1, c_2, \dots$ separated by commas.)

12. (1 pt) ttulib/M2360/chap2/leon.2.2/n07.pg

(S. 2-2, cf. no. 7))

Let A and B be 3×3 matrices with $\det(A) = 2$ and $\det(B) = 7$.

Find the value of:

(a) $\det(AB) = \underline{\hspace{2cm}}$

(b) $\det(2A) = \underline{\hspace{2cm}}$

(c) $\det(3AB) = \underline{\hspace{2cm}}$

(d) $\det(AB^{-1}) = \underline{\hspace{2cm}}$

13. (1 pt) ttulib/M2360/chap2/leon.2.2/ur_la_6.1.pg

Consider the following general matrix equation:

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

which can also be abbreviated as:

$$A = MX$$

By definition, the determinant of M is given by

$$\det(M) = m_{11}m_{22} - m_{12}m_{21}$$

The following questions are about the relationship between the determinant of M and the ability to solve the equation above for A in terms of X or for X in terms of A .

Check the boxes which make the statement correct:

If the $\det(M) \neq 0$ then

- A. given any A there is one and only one X which will satisfy the equation.
- B. some values of X will have no values of A which satisfy the equation.
- C. some values of A will have no values of X which will satisfy the equation.
- D. some values of A (such as $A = 0$) will allow more than one X to satisfy the equation.
- E. given any X there is one and only one A which will satisfy the equation.
- F. some values of X will have more than one value of A which satisfy the equation.

Check the boxes which make the statement correct:

If the $\det(M) = 0$ then

- A. some values of A (such as $A = 0$) will allow more than one X to satisfy the equation.
- B. some values of A will have no values of X which will satisfy the equation.
- C. given any X there is one and only one A which will satisfy the equation.
- D. there is no value of X which satisfies the equation when $A = 0$.
- E. given any A there is one and only one X which will satisfy the equation.

Check the conditions that guarantee that $\det(M) = 0$:

- A. There is some value of A for which no value of X satisfies the equation.
- B. Given any X there is one and only one A which will satisfy the equation.
- C. When $A = 0$ there is more than one X which satisfies the equation.
- D. Given any A there is one and only one X which will satisfy the equation.

14. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.9.pg

If A and B are 3×3 matrices, $\det(A) = 3$, $\det(B) = -7$, then

$\det(AB) = \underline{\hspace{2cm}}$,

$\det(2A) = \underline{\hspace{2cm}}$,

$\det(A^T) = \underline{\hspace{2cm}}$,

$\det(B^{-1}) = \underline{\hspace{2cm}}$,

$\det(B^4) = \underline{\hspace{2cm}}$.

15. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.11.pg

Find the determinant of the matrix

$$M = \begin{bmatrix} -3 & -8 & 1 \\ 0 & -5 & -1 \\ 0 & 0 & 5 \end{bmatrix}.$$

$\det(M) = \underline{\hspace{2cm}}$.

16. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.12.pg

A square matrix is called a permutation matrix if each row and each column contains exactly one entry 1, with all other entries being 0. An example is

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Find the determinant of this matrix.

$\det(P) = \underline{\hspace{2cm}}$.

17. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.13.pg

Find the determinant of the matrix

$$A = \begin{bmatrix} -3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & -8 & 5 & 0 \\ 1 & -9 & 4 & 5 \end{bmatrix}.$$

$\det(A) = \underline{\hspace{2cm}}$.

18. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.15.pg

Find the determinant of the matrix

$$M = \begin{bmatrix} 3 & 0 & 0 & -2 & 0 \\ -1 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & -3 & -2 \\ 0 & -1 & 3 & 0 & 0 \end{bmatrix}.$$

$\det(M) = \underline{\hspace{2cm}}$.

19. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.16.pg

Find the determinant of the $n \times n$ matrix A with 2's on the diagonal, 1's above the diagonal, and 0's below the diagonal.

$\det(A) = \underline{\hspace{2cm}}$.

20. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.17.pg

If the determinant of a 5×5 matrix A is $\det(A) = 10$, and the matrix B is obtained from A by multiplying the second row by 7, then $\det(B) = \underline{\hspace{2cm}}$.

21. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.18.pg

If the determinant of a 4×4 matrix A is $\det(A) = 4$, and the matrix C is obtained from A by swapping the second and third columns, then $\det(C) = \underline{\hspace{2cm}}$.

22. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.19.pg

If the determinant of a 5×5 matrix A is $\det(A) = 9$, and the matrix D is obtained from A by adding 10 times the third row to the second, then $\det(D) = \underline{\hspace{2cm}}$.

23. (1 pt) ttulib/M2360/chap2/leon.2.2/ur.la.6.24.pg

Let $A = \begin{bmatrix} 1 & 3 \\ 6 & 7 \end{bmatrix}$.

Find the following:

(a) $\det(A) = \underline{\hspace{2cm}}$,

(b) the matrix of cofactors $C = \begin{bmatrix} \underline{\hspace{2cm}} & \underline{\hspace{2cm}} \\ \underline{\hspace{2cm}} & \underline{\hspace{2cm}} \end{bmatrix}$,

Determine the nullspace of each of the following matrices. Write your answer as a linear combination of vectors as in example 9 (p. 137). E.g., if the null space is the set of all vectors in the form

$$\alpha \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}$$

where α, β are scalars, you would enter

$$a * < 1, 2, 3 > + b * < 3, 0, -1 >$$

If the null space is the trivial subspace $\{0\}$, just enter the zero vector $<0, \dots, 0>$ (use the appropriate number of 0's.)

$$(a) A = \begin{bmatrix} 1 & 3 \\ 4 & 13 \end{bmatrix}, \quad N(A) = \{ \text{_____} \}$$

$$(b) A = \begin{bmatrix} 1 & 1 & 2 & -3 \\ 3 & 1 & 1 & -2 \end{bmatrix}, \quad N(A) = \{ \text{_____} \}$$

$$(c) A = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 2 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad N(A) = \{ \text{_____} \}$$

32. (1 pt) ttulib/M2360/chap3/leon.3.2/n09.pg

(S. 3-2, cf. no. 9)

Determine whether the following are spanning sets for R^2 :

$$(a) \{(1, 2)^T, (1, 3)^T\}$$

- yes
- no

$$(b) \{(2, 5)^T, (-6, -15)^T\}$$

- yes
- no

$$(c) \{(-2, 1)^T, (2, -1)^T, (1, 1)^T\}$$

- yes
- no

$$(d) \{(1, -2)^T, (-1, 2)^T, (-5, 10)^T\}$$

- yes
- no

33. (1 pt) ttulib/M2360/chap3/leon.3.2/n10.pg

(S. 3-2, cf. no. 10)

Determine whether the following are spanning sets for R^3 :

$$(a) \{(1, 2, 1)^T, (1, 3, 1)^T, (1, 1, 1)^T\}$$

- yes
- no

$$(b) \{(2, 5, 2)^T, (-6, -15, 1)^T\}$$

- yes
- no

$$(c) \{(-2, 1, 0)^T, (2, -1, 1)^T, (2, 1, 3)^T\}$$

- yes
- no

$$(d) \{(1, 7, 2)^T, (-1, 1, 7)^T, (2, -3, 4)^T\}$$

- yes
- no

34. (1 pt) ttulib/M2360/chap3/leon.3.2/n11b.pg

(S. 3-2,)

Demonstrate that $\{(1, 1)^T, (1, 3)^T\}$ is a spanning set for R^2 by showing how to express an arbitrary member $\begin{pmatrix} a \\ b \end{pmatrix}$ of R^2 as a linear combination of its members. (Your answers will be an expression in a, b .)

$$\begin{pmatrix} a \\ b \end{pmatrix} = \text{_____} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \text{_____} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

35. (1 pt) ttulib/M2360/chap3/leon.3.2/n14.pg

(S. 3-2, cf. no. 14)

Which of the following are spanning sets for P_3 (the set of polynomials of degree less than 3)?

$$(a) \{x, x^2 - x, x^2 + x\}$$

- yes
- no

$$(b) \{x + 1, x - 1, x^2 + x\}$$

- yes
- no

$$(c) \{x + 1, x^2 - 1\}$$

- yes
- no

36. (1 pt) ttulib/M2360/chap3/leon.3.2/ur.la.8.1.pg

Which of the following subsets of P_2 are subspaces of P_2 ?

- A. $\{p(t) \mid p(6) = 0\}$
- B. $\{p(t) \mid p'(t) + 5p(t) + 8 = 0\}$
- C. $\{p(t) \mid p(-t) = -p(t) \text{ for all } t\}$
- D. $\{p(t) \mid p'(t) \text{ is constant}\}$
- E. $\{p(t) \mid p'(0) = p(3)\}$
- F. $\{p(t) \mid p(2) = 2\}$

37. (1 pt) ttulib/M2360/chap3/leon.3.2/ur.la.8.2.dsg.pg

Which of the following subsets of $\mathbb{R}^{3 \times 3}$ are subspaces of $\mathbb{R}^{3 \times 3}$?

- A. The 3×3 matrices A such that the vector $\begin{pmatrix} 6 \\ 3 \\ 4 \end{pmatrix}$ is in the null space of A
- B. The 3×3 matrices with all zeros in the second row
- C. The 3×3 matrices with determinant 1
- D. The invertible 3×3 matrices
- E. The 3×3 matrices with determinant 0

- F. The diagonal 3×3 matrices

38. (1 pt) ttulib/M2360/chap3/leon.3.3/n01.pg

(S. 3-3, cf. no. 1)

Determine whether the following vectors are linearly independent in R^2 :

- (a) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}$
- yes
 - no
- (b) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}, \begin{pmatrix} -6 \\ -4 \end{pmatrix}$
- yes
 - no
- (c) $\begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 7 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- yes
 - no
- (d) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- yes
 - no

39. (1 pt) ttulib/M2360/chap3/leon.3.3/n02.pg

(S. 3-3, cf. no. 2)

Determine whether the following vectors are linearly independent in R^3 :

- (a) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
- yes
 - no
- (b) $\begin{pmatrix} 3 \\ 2 \\ 7 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \\ -4 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$
- yes

- no

- (c) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix}$
- yes
 - no

- (d) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ -7 \\ 3 \end{pmatrix}, \begin{pmatrix} 8 \\ 2 \\ -5 \end{pmatrix}$
- yes
 - no

- (e) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ -7 \\ 3 \end{pmatrix},$
- yes
 - no

40. (1 pt) ttulib/M2360/chap3/leon.3.3/n06.pg

(S. 3-3, cf. no. 6)

Determine whether the following vectors (polynomials) are linearly independent in P_3 :

- (a) $x, x^2, x^2 - 2$
- yes
 - no
- (b) $1 + x, 1 - x, 1 - x^2, x^2$
- yes
 - no
- (c) $1 + x, 1 - x, 1 - x^2$
- yes
 - no
- (d) $x - 2, x + 1, 2x^2 + x$
- yes
 - no